Mathematical Notes

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1 Introduction

Definition 1.1: Holomorfic function

A function $f\colon \mathbb{C} \supseteq \Omega \to \mathbb{C}$ is *holomorphic* if it is complex differentiable on Ω, meaning that for any $z \in \Omega$

$$f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h}$$

exists. Equivalently,

$$f(z+h) = f(z) + hf'(z) + o(h).$$

Equivalently, f is differentiable as a function $\mathbb{R} \to \mathbb{R}$ and its differential is a matrix of multiplication by a complex number

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

for some reals a, b. Writing this using partials, we get

Definition 1.2: Cauchy - Riemann equations

Let f be holomorphic and write $f(x+iy) = u(x,y) + iv(x,y)$. Then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

TODO: $\frac{\partial}{\partial z}$, $\frac{\partial}{\partial \bar{z}}$.

Lemma 1.3: Closedness of fdz

Let $dz = dx + idy$. Then f is holomorphic iff the form fdz is closed, meaning

$$d(fdz) = 0.$$

All arithmetic properties are the same as for reals.

Example 1.4: Some holomorphic functions

Any polynomial, convergent power series (not proven yet), exp, all trigonometric functions outside their poles. sin and cos are unbounded for large y and tan is bounded for large y.

2 Extremes, Analytic functions, Series

2.1 Extremes

Theorem 2.1: Extremes of holomorphic functions

Ω connected and bounded, f holomorphic on Ω. Then $|f|$ does not have local extremes unless it is constant.

For nondegenerate extremes. $f(z) = u(x,y) + iv(x,y)$. $|f|^2 = f \cdot \overline{f} = u^2 + v^2 = g$. $Dg = 0$ and $D^2 g$ can't be negatively-definite.

$$g_x = \frac{\partial}{\partial x}(u^2 + v^2) = 2uu_x + 2vv_x$$

$$g_y = \frac{\partial}{\partial y}(u^2 + v^2) = 2uu_y + 2vv_y.$$

If $(g_x, g_y) = (0,0)$ then $(u,v) = 0$ or from Cauchy Riemann

$$\det \begin{bmatrix} u_x & v_x \\ u_y & v_y \end{bmatrix} = 0 \implies u_x = v_x = u_y = v_y = 0 \implies f'(x) = 0$$

$$\operatorname{tr} D^2 g = g_{xx} + g_{yy} = 2u(u_{xx} + u_{uu}) + 2v(v_{xx} + v_{yy}) = 0,$$

which is a contradiction. □

Proof. Take a circle γ around the point maximizing $|f|$:

$$f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(w)}{w - z_0} dw = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{it})$$

This is the average of f along γ, and we get

$$|f(z_0)| \leqslant \max_{z \in \gamma} |f(z)|.$$

But from maximality, $|f|$ must be constantly equal $|f(z_0)|$ on γ. Analogously, it is constant inside any circle (contained in Ω) around z_0. By some standard topology i is constant everywhere as Ω is open and connected.

□

2.2 Analytic functions and Series

Theorem 2.2: Holomorphic $\implies$ Analytic

Any holomorphic function is analytic.

Proof. Take a circle γ around z_0, $w \in \gamma$. For z in the interior of γ (to make the series converge), we have

$$\frac{1}{w-z} = \frac{1}{w-z_0} \sum_{n=0}^{\infty} \left(\frac{z-z_0}{w-z_0} \right)^n,$$

hence

$$f(z) = \frac{1}{2\pi i} \int_\gamma f(w) \sum_{n=0}^{\infty} \frac{(z-z_0)^n}{(w-z_0)^{n+1}} dw = \sum \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n.$$

We can exchange integration with summation, since the inside is uniformly convergent on a compact set. □

Corollary: if f is analytic around 0, then

$$f(w) = \sum a_n (z-z_0)^n$$

where $|a_n| < \frac{M}{R^n}$ for an upper bound M of $|f|$ inside the considered disk. Note that R can be taken to be maximal possible. Consider arctan. Its convergence radius at 0 is 1, so it can't extend to an entire function. In this case, it has poles at $\pm i$.

Theorem 2.3: Analytic $\implies$ Holomorphic

Any analytic function is holomorphic.

Proof. This is a standard bound-bash. The same argument as for real analytic functions. □

Lemma 2.4: Hadamard lemma

$f \colon \mathbb{R} \to \mathbb{R}$. $f(0) = 0$. If $f \in C^k$, then $\exists g \in C^{k-1}$ such that $f(x) = xg(x)$.

Proof.

$$f(x) = \int_0^1 \frac{d}{dt} f(xt) dt = \int_0^1 x f'(xt) dt = x \int_0^1 f'(xt) dt$$

□

Theorem 2.5: degree of vanishing

f can be uniquely written as $z^n g(z)$ where $n \geqslant 0$.